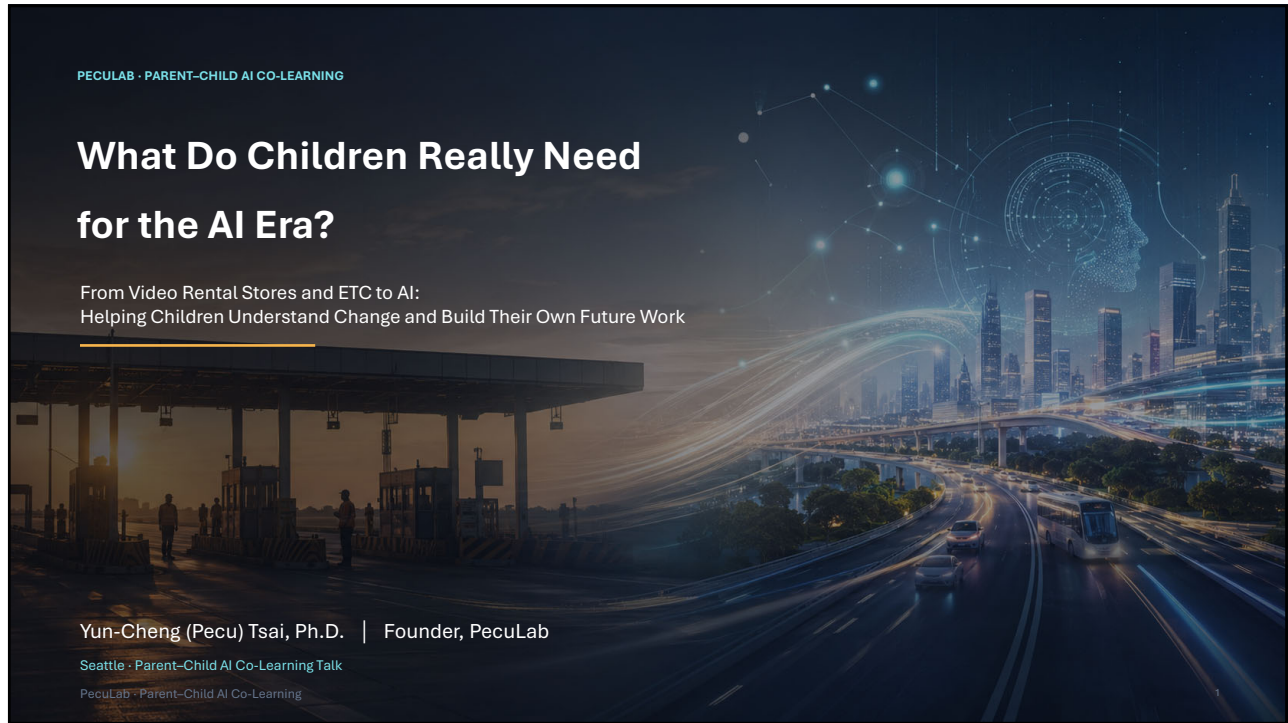


PECULAB · PARENT-CHILD AI CO-LEARNING

What Do Children Really Need for the AI Era?

From Video Rental Stores and ETC to AI:
Helping Children Understand Change and Build Their Own Future Work



Yun-Cheng (Pecu) Tsai, Ph.D. | Founder, Peculab

Seattle · Parent-Child AI Co-Learning Talk

PecuLab · Parent-Child AI Co-Learning

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WHO I AM

Why I Started This Small Co-Learning Club

Bringing observations from AI, education, and the real world
into skills children can begin practicing today.

**Yun-Cheng (Pecu)
Tsai, Ph.D.**

AI Scholar &
Applied Intelligence Educator

Founder, Peculab
Former Associate Professor, National Taiwan
Normal University
AI · Educational Technology · Data Analytics
Project-Based Learning & Social Applications

I am not bringing university lectures to children. I want to help them notice problems, ask better questions, use AI wisely and turn ideas into work they can share.

“ I hope to translate “what is happening in the AI world” into abilities that children and families can begin practicing today.

PecuLab: a small, project-centered AI co-learning lab built around relationships and feedback.

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CONTEXT, NOT A SALES PITCH

My work is about moving AI from concepts into cities, communities, and the real world.

Future Intelligence Hub

AI-powered city innovation infrastructure

Seattle: Global HQ and Command Center

Managed by ESGIN

Global ESG Leadership Organization

Implementation-oriented sustainability platform

Why it matters here

From global questions about AI, youth, and social innovation to abilities children can begin practicing today.

SCAN ME



Large systems think about the future of cities; PecuLab starts with one question from a child.

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TODAY

What Will We Practice Together Today?

Understand change, imagine the future, and help children turn curiosity into work they can share.

01

Hear a story from video rental stores and ETC to AI

02

Choose something familiar and imagine a new solution for 2036

03

Understand how AI can become a partner for thinking and creating

Warm-up question: What ability do you most hope your child keeps in the AI era?

Creativity

Judgment

Expression

Problem Solving

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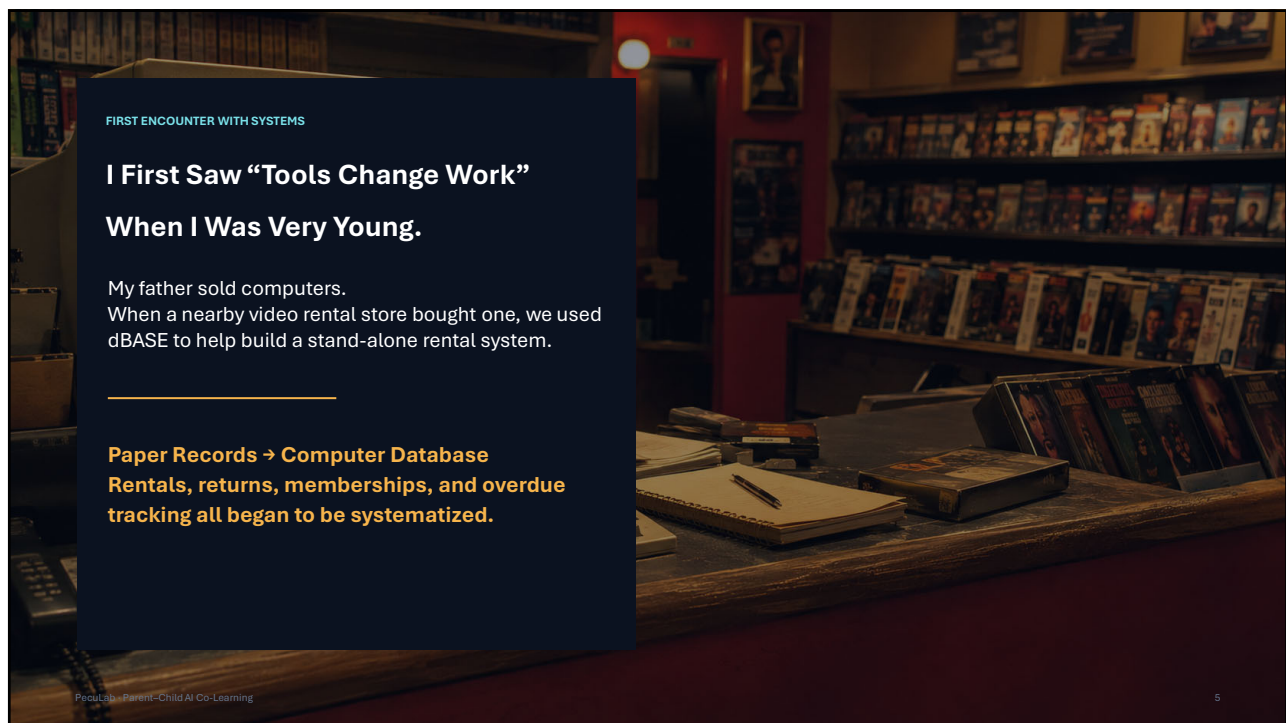
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PATTERN

From dBASE Video Rental Stores, to ETC, to AI Today

Tools keep changing. What accumulates is the ability to understand processes, learn new tools, and propose new solutions.

As a Child

dBASE Video Rental System

Paper records were replaced by a stand-alone database

At 22

ETC Audit Software Development

Manual toll collection was redesigned by automation

Today

AI Enters Education and Work

Content, decisions, learning, and services are being redesigned

I do not chase tools blindly. I care more about:
What process is this tool changing? What problem does it solve?
What new possibilities does it create?

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A TURNING POINT

At 22, I Stood at Another Turning Point.

I joined the audit software development team for Taiwan's highway ETC toll collection system.

**Technology is not the problem.
But when the world turns too quickly, people may not be ready.**

The point is not to predict which jobs will disappear. It is to learn to see changing needs and new solutions.

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A SIMPLE LENS

What Disappears Is Not the Need — It Is the Old Solution.

Video Rental Store

Rentals · Returns · Overdues ·
Membership Records

**People still need:
Stories, entertainment,
companionship, and rest.**

Streaming & Interactive Content

Recommendations · On-demand access ·
Shared viewing · Immersive experiences

**The form changed.
Human needs did not.**

Key Capabilities

**See the need
See the shift
Design the next
solution**

The same pattern appears in toll booths, bank branches, tutoring, delivery, healthcare, and the future of education.

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PARENT-CHILD WARM-UP

Why Did Video Rental Stores Disappear?

Families: discuss together for 1 minute.

1. What did it appear to provide?
2. What did people truly need?
3. How do we meet that need today?

Entertainment • Companionship • Choice • Shared memories • Rest

The need to rent a tape did not disappear; the ways people enjoy stories and companionship changed.

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DESIGN TOGETHER

Parent-Child Future Design: What Could Change by 2036?

We do not need to predict correctly.
We are practicing how to see enduring needs and possible new solutions.

Tutoring	Food Delivery	Convenience Stores	Healthcare Visits
Buying Clothes	Driving	Banking	Travel

1. What need does it meet today?
2. What feels inconvenient today?
3. What might AI change?
4. What could its 2036 version be?

This is not prediction. It is practice:
first understand, then redesign.

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DESIGN THINKING

Do Not Chase Tools. Use Tools to Solve Problems.

Children observe and choose;
AI helps explore, organize, simulate, and create a first version.

01

Notice Change

Find an everyday situation
that may be rewritten

02

Find the Enduring Need

Ask: What do people really
need?

03

Imagine the Future

Imagine new 2036
scenarios

04

Design a Solution

Choose a scenario and turn it
into a service or project

05

Build a Prototype

Turn the idea into a small
working artifact

**This process is not about prompt tricks.
It builds problem awareness, judgment, creativity, and expression.**

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AI AS A THOUGHT PARTNER

AI Should Not Help Children Skip Thinking. AI Should Help Them Take Their Thinking Further.

AI Can Help With

Seeing different user perspectives
Organizing information and questions
Simulating future scenarios
Creating a first version

Children Still Need To

Choose worthwhile problems
Question AI's answers
Make design decisions
Take responsibility for impact and ethics

**AI generates candidate answers.
People need to practice judgment and choice.**

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LIVE DEMO

Let AI Expand Our Thinking — Not Decide the Answer for Us.

Demo Prompt:

We are imagining tutoring in 2036.

Please help us analyze:

1. What do children and parents truly need?
 2. What are common difficulties in tutoring today?
 3. Propose three different future learning scenarios.
- But do not decide which one is best for us.

Every time we use AI,
we ask three questions:

- ① What new perspective did it give me?
- ② What do I not fully agree with?
- ③ What do I want to change in the end?

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TAKE-HOME FRAMEWORK

Parents Do Not Need To Be AI Experts To Learn Alongside Their Children.

You do not need to teach the answer.

You only need to help your child ask a deeper question.

1 What problem do you notice?

Start with everyday life

2 Who is truly affected?

Practice empathy for different roles

3 Do you agree with AI's answer?

Practice judgment and verification

4 How would you redesign it?

Practice creating and expressing

For younger children: ask, draw, and talk together.
Do not rush to require a complete project.

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FROM TALK TO WORK

What Summer Can Leave Behind Is Not Just a Class — It Is a Project.

Help children turn ideas into work they can show, share, and revise.

Family Storybook or Picture Book

Combine family memories, children’s imagination, and AI-supported words and images

A Small Project Around a Child’s Interest

Create a research poster about dinosaurs, space, animals, music, sports, or another interest

AI-Supported Presentation or Research Poster

Learn to research, organize ideas, and articulate personal judgment

Everyday Problem-Solving Proposal

Design a solution for one small problem at school, in the community, or at home

The project is not about looking polished.
It helps children practice: What did I notice? What do I think?
How did I use AI? Why did I make these choices?

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PECULAB

From One Talk To a Small Co-Learning Club.

Starting with lecture families and trusted referrals.
So every child can be seen, receive feedback,
and leave with work of their own.

Parent-Child AI Co-Learning Studio

K-3 · 4 weeks · Parent participation
Create a story or everyday-life project together

Youth Project Lab

Grades 4-8 · 8 weeks · Small-group projects
Create an AI social-impact project and present at Demo Day

Small by design. Deep by intention.

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THE INVITATION

You Do Not Need To Choose Your Child’s Path Today. Start by helping them become someone who can understand problems and create value when the world changes.

PecuLab Founding Families

The first season is open only to lecture families and trusted referrals.
I will recommend the most suitable co-learning path based on your child’s age,
interests, and preferred way to participate.



PecuLab · AI Co-Learning Lab

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FIH

Future Intelligence Hub

AI-powered city innovation infrastructure

ModelESGINCitiesEventsProgramsHackathonImpactPartner

DOHA SOLUTIONS PLATFORM INITIATIVES

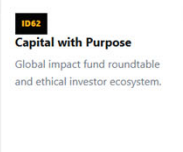
FIH supports ESGIN initiatives that move from global recognition into city-level execution.



ID169

Leader for Young Leaders

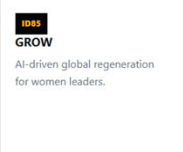
A world leaders mentorship initiative for youth development.



ID162

Capital with Purpose

Global impact fund roundtable and ethical investor ecosystem.



ID185

GROW

AI-driven global regeneration for women leaders.



ID167

Responsible AI for Business

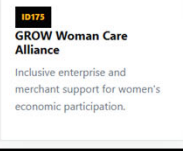
AI-driven ESG and SDG transformation for SMEs, enterprises, and communities.



ID198

Youth and Women Financial Literacy

Economic empowerment through inclusive financial literacy.



ID175

GROW Woman Care Alliance

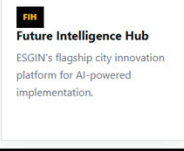
Inclusive enterprise and merchant support for women's economic participation.



ID171

Responsible Digital for Youth

Ethical, responsible, and safe digital citizenship for young people.



FIH

Future Intelligence Hub

ESGIN's flagship city innovation platform for AI-powered implementation.

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